

Transformations with Multiple Parameters

Precalculus Functions and Composition

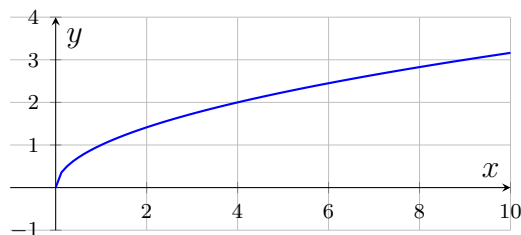
Transformations from a table, parameters inside an equation, and the order of composition

Directions: Every problem here has *more than one* parameter acting at once. Name each parameter before you compute, and keep the inside-the-function moves separate from the outside-the-function moves.

The parameter map — read every problem against this

$$g(x) = \underbrace{a}_{\substack{\text{vertical stretch} \\ \text{reflect if } a < 0}} \cdot f\left(\underbrace{b}_{\substack{\text{horizontal} \\ \text{compress by } 1/b}} (x - \underbrace{h}_{\substack{\text{shift right} \\ h \text{ units}}})\right) + \underbrace{k}_{\substack{\text{shift up} \\ k \text{ units}}}$$

Outside the function (a, k) acts on the *output* and does what it says. **Inside** (b, h) acts on the *input* and does the opposite of what it looks like. The inside must be *factored* into the form $b(x - h)$ before h can be read: in $f(2x - 6) = f(2(x - 3))$ the shift is 3, not 6.



The parent function $f(x) = \sqrt{x}$.

1. Let $f(x) = x^2$ and $g(x) = 2f(x - 3) - 5$. Fill in the chain, then state $g(5)$.

x	$x - 3$	$f(x - 3)$	$2f(x - 3)$	$2f(x - 3) - 5$
5				

Answer: _____

2. Let $f(x) = x^2$ and $h(x) = -f(x + 1) + 4$.

Which two parameters are non-trivial here, and what does each do? Then find $h(2)$.

Answer: _____

3. Let $g(x) = 3\sqrt{x - 4} + 1$.

Solve $g(x) = 7$. Undo the outside parameters first, then the inside one, and say which step each parameter is undone in.

Answer: _____

4. Let $y = 4 \sin\left(2\left(x - \frac{\pi}{6}\right)\right) + 1$.

Identify the amplitude, the period, the phase shift and the midline. Then find y when $x = \frac{5\pi}{12}$.

Answer: _____

5. Two transformation maps: $s(x) = x + 4$ (a shift) and $d(x) = 2x$ (a stretch).

Write a simplified formula for $(d \circ s)(x) = d(s(x))$, and state in words which move happens first.

Answer: _____

6. Using the same $s(x) = x + 4$ and $d(x) = 2x$:

For what value of x does $(d \circ s)(x) = 20$? Then explain in one sentence why $(s \circ d)(x)$ would *not* give the same answer.

Answer: _____

7. Rewrite $y = 3 \sin\left(2x - \frac{\pi}{3}\right)$ in the form $A \sin(B(x - C))$.

State the phase shift C , and explain why it is not $\frac{\pi}{3}$.

Answer: _____

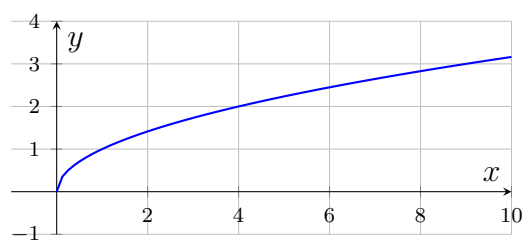
- 8.** Let $f(x) = \sqrt{x}$ (graphed above) and $g(x) = -2f(3(x-1)) + 5$. List all four parameters and what each one does, then compute $g(4)$.

Answer: _____

- 9.** Let $g(x) = 2\sqrt{x+3} - 4$.

Find the x -intercept of g . (Set the output to zero and undo the parameters in the reverse of the order that builds them.)

Answer: _____



The parent function $f(x) = \sqrt{x}$.

- 10.** Using the parent $f(x) = \sqrt{x}$ graphed above, sketch $y = -2f\left(\frac{x}{2} + 1\right) + 3$ on the grid below.

Before you sketch, factor the inside expression and list the four transformations in an order that produces the correct graph. Then explain why the horizontal shift is **2 units left** even though the inside reads “+1”. Label the image of the parent’s starting point on your sketch.

